WESTFIELD CASE

PART 2

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Recap: Descriptive Analytics

- Last lecture revealed the current state of WestField. Here is what the data told us:
  - **Demand is outpacing capacity.** Referral rates have grown steadily year over year, resulting in a waitlist of 5000+ children
  - **Ongoing care dominates available capacity.** The majority of all appointments go to active clients already in treatment, who have varying treatment intensity needs.
  - **~20% of capacity is silently wasted**. Roughly one in five scheduled appointments goes uncompleted due to no-shows and cancellations.
  - **Aging-out adds urgency to every delay.** Every week spent waiting is a week closer to aging out of eligibility entirely, making timely access not just a service quality issue but a matter of whether a child receives any care at all.

WHAT WILL HAPPEN?

Predictive Analytics

Next Step: Predictive Analytics

- Descriptive analytics told us where we are. Predictive analytics tells us where we are headed

Descriptive *"What happened?"*	**Example:** Waitlist length and wait-time distributions, cancellation rates
Diagnostic *"What factors caused it to happen?"*	**Example:** treatment length variability, eligibility policies
Predictive *"What will happen?"*	**Example:** Time-series forecasting, Regression models, Machine-learning models
Prescriptive *"What should we do?"*	**Example:** Capacity allocation models & scheduling optimization

What Do We Want to Predict?

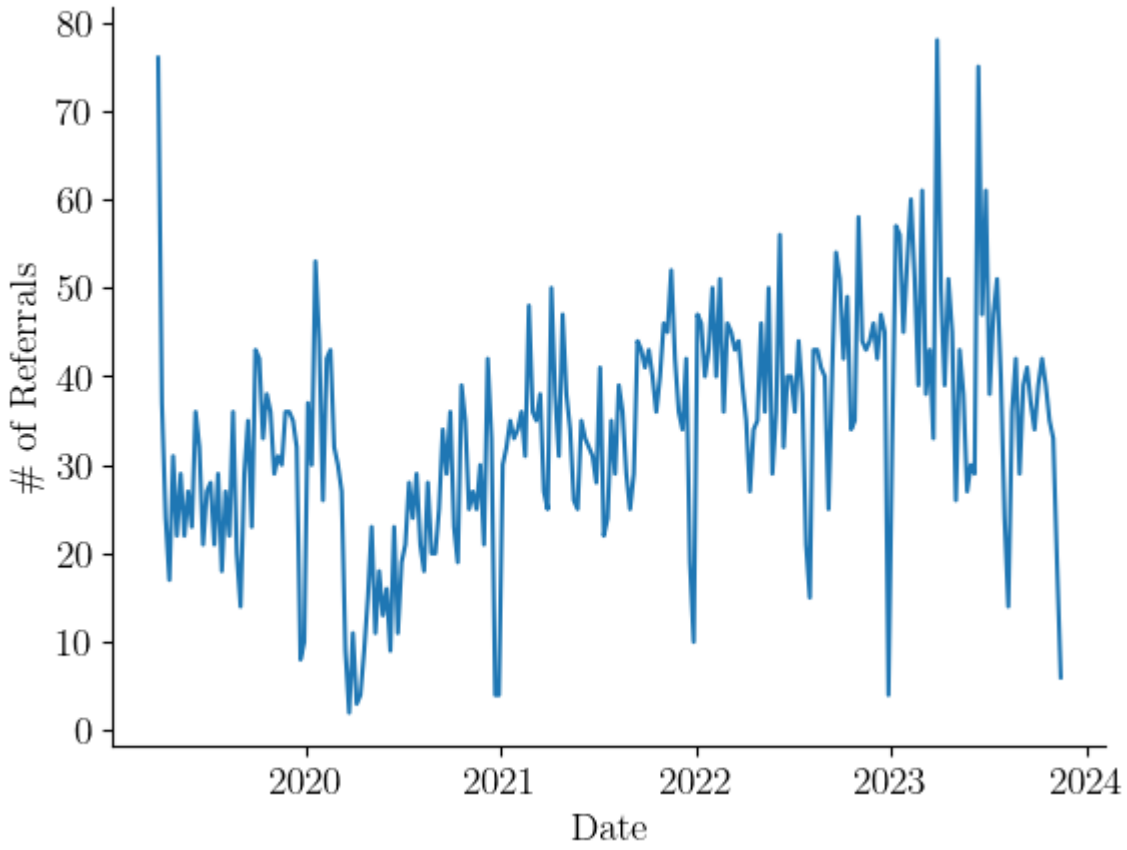
- Each issue identified in descriptive analytics points directly to something we need to forecast
  1. Future Demand/Referral Rate in the coming weeks/months
  2. The volume of children who will near aging-out on the waitlist throughout the year
  3. Which specific appointments are at risk of being cancelled?
  4. How many appointments will a given patient need to complete their episode?

PREDICTING: FUTURE DEMAND

Using Time-Series Forecasting with ARIMA

Predicting Future Demand

- Processes like arrival rates form time series data
  - Defined as: *a series of values obtained at successive times, often with equal intervals between observations*
- Predicting future values = **forecasting**
- Accurate predictions can help inform planning decisions around resources needed to accommodate demand spikes



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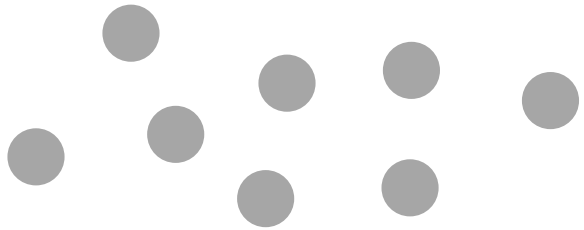
Forecasting Models

- Forecasting uses models specifically designed for sequential, time-dependent data (unlike single one-off predictions)

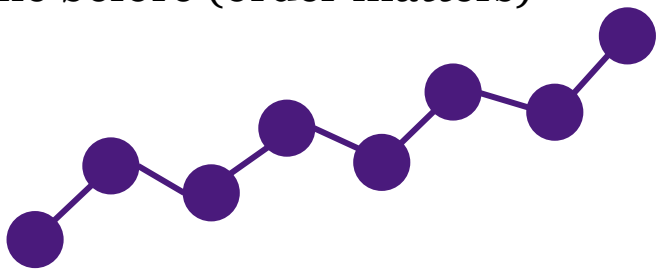
i.i.d. data

Each observation is independent and order doesn't matter (e.g., coin flips)



Time series (dependent data)

Each observation depends on what came before (order matters)



- We can forecast time-series using **ARIMA**
  - ARIMA models refer to time series processes parameterized by Autoregressive (AR), Integrated (I), and Moving Average (MA) terms

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ARIMA

- ARIMA has three parts. Each solves a specific area with time series data.

AR Auto-Regressive

The past predicts the future

If referrals were high last week, they will probably be high this week. The AR part uses your own recent history as input.

$$Y_t = \beta + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p}$$

p = how many past weeks to look back

Integrated (Differencing)

Remove the upward drift first

If referrals were slowly climbing each year, we subtract consecutive values to get a flat signal. Now the model can focus on up-and-down variation.

$$Z_t = Y_{t+1} - Y_t$$

d = how many times to difference

MA Moving Average

Learn from past mistakes

If the model over-predicted last week, it adjusts downward this week. The MA part feeds past forecast errors back in as a self-correction.

$$Y_t = \beta + \omega_1 \varepsilon_{t-1} + \omega_2 \varepsilon_{t-2} + \ldots + \omega_q \varepsilon_{t-q}$$

q = how many past errors to include

Together: ARIMA (p, d, q). Each of p, d, q is a number you choose.

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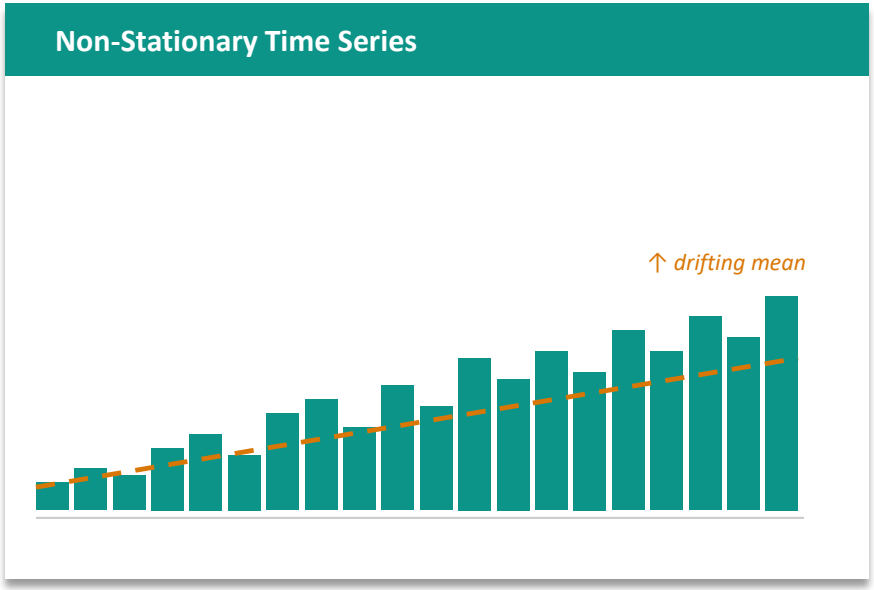
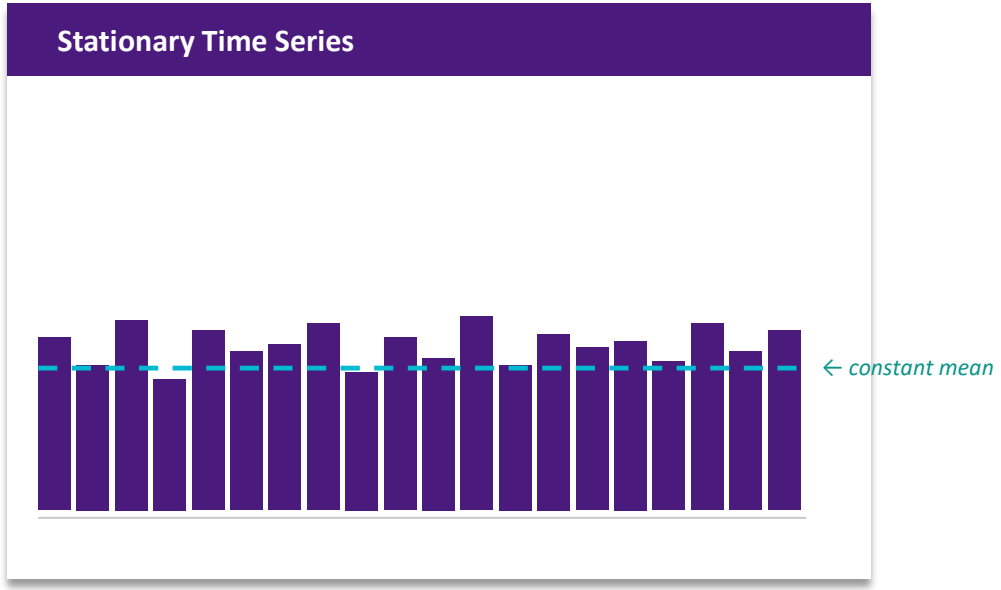
The 'I' in ARIMA: Removing the Trend First

- Before the AR and MA parts can work, the data needs a stable average
- **Stationarity** refers to the behaviour of the **mean** of a time series
  - A stationary time series has a constant mean
  - A non-stationary time series is one in which the mean changes over time (e.g., values decrease or increase on average as times progress)





The 'I' in ARIMA: Removing the Trend First

- **Integration (I)**:
  - Can handle non-stationarity in a time series
  - Often, a non-stationary time series can be made stationary by taking the difference between successive observations

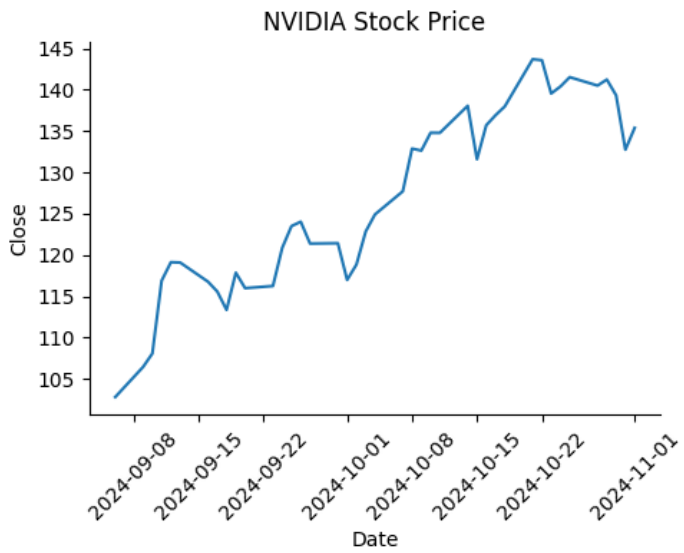
$$\mathbf{Z_t = Y_{t+1} - Y_t}$$

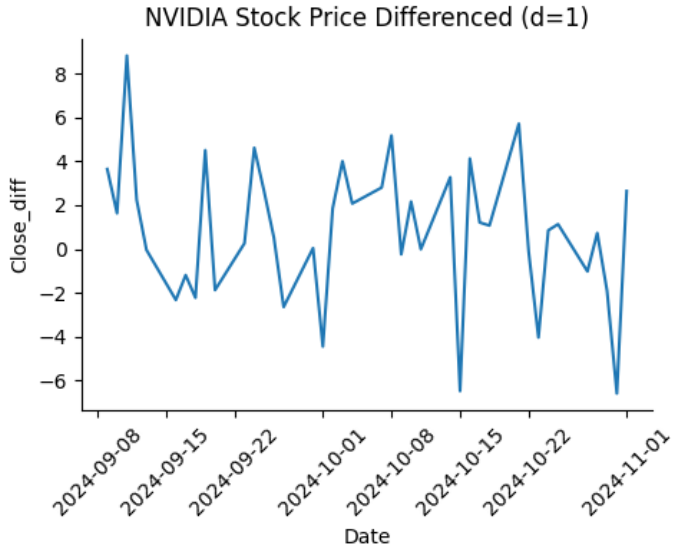
  - Now, instead of modeling time series Y, we model Z
  - The order of differencing **(d)** refers to how many times we successively difference observations

$$Z_t = Y_{t+1} - Y_t,\ d = 1$$

$$Q_t = Z_{t+1} - Z_t,\ d = 2$$

- Examples: Stock prices
  - They typically exhibit non-stationarity which is why people tend to predict returns (i.e., differences between successive day's prices instead of the absolute price)





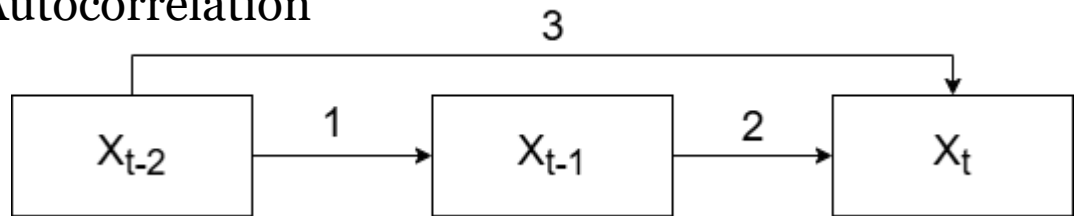
Fitting AR and MA parameters

- Auto-Regressive (AR): $\mathbf{Y_t = \beta + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p}}$
  - This week's referrals = a baseline + a portion of last week's count + a smaller portion from two weeks ago + …
  - The ϕ values are fitted from data. They tell you how much each past week matters.
- Moving Average: $\mathbf{Y_t = \beta + \omega_1 \varepsilon_{t-1} + \omega_2 \varepsilon_{t-2} + \ldots + \omega_q \varepsilon_{t-q}}$
  - This week's prediction = baseline + a correction based on how wrong last week was + how wrong two weeks ago was + …
  - ε = forecast error; ω = how much each error counts
- Both parameters are fitted simultaneously using **maximum likelihood**

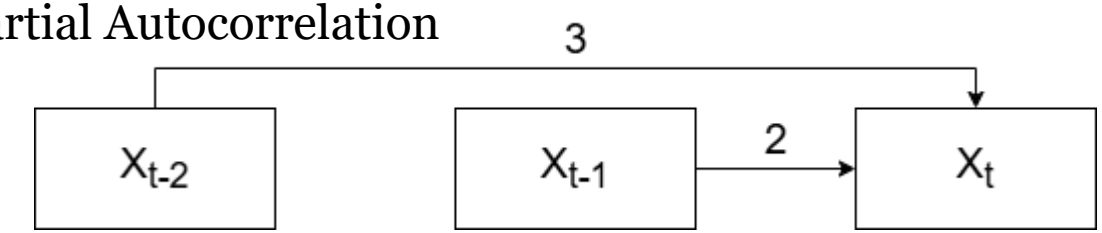
Types of Data Correlation

- Autocorrelation (ACF): direct + indirect effects
- Partial autocorrelation (PACF): direct effects
- Example: Suppose today's temperature is correlated with temperatures from the past 7 days
  - ACF will show high correlation at lag 1, lag 2, lag 3… all the way to lag 7, slowly fading. It looks like every past day matters, but usually less the farther away it was
  - PACF might show a spike only at lag 1, then nothing. This tells you: *only yesterday directly drives today — the rest is just a chain reaction passing through yesterday*
  - PACF might show you that instead of a complex model using 7 days of history, you may only need 1 day

Autocorrelation



Partial Autocorrelation

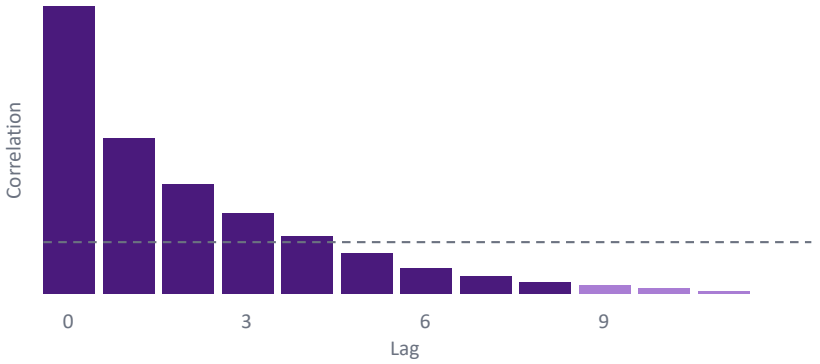


How Do We Choose p and q?

- We can use plots of autocorrelation and partial autocorrelation to determine q and p, respectively
- This is a visual diagnostics method to determine these values (but we can also use AIC grid)

ACF — Autocorrelation

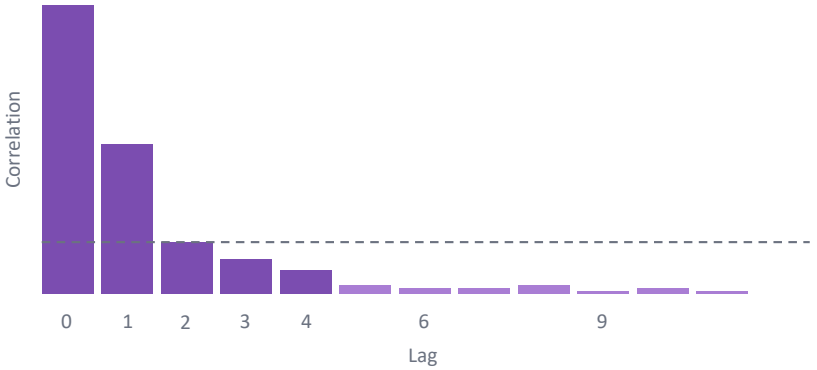
Correlation between this week and each past week — including indirect links.



Slow decay → AR process present. Use ACF to guide q (MA order).

PACF — Partial Autocorrelation

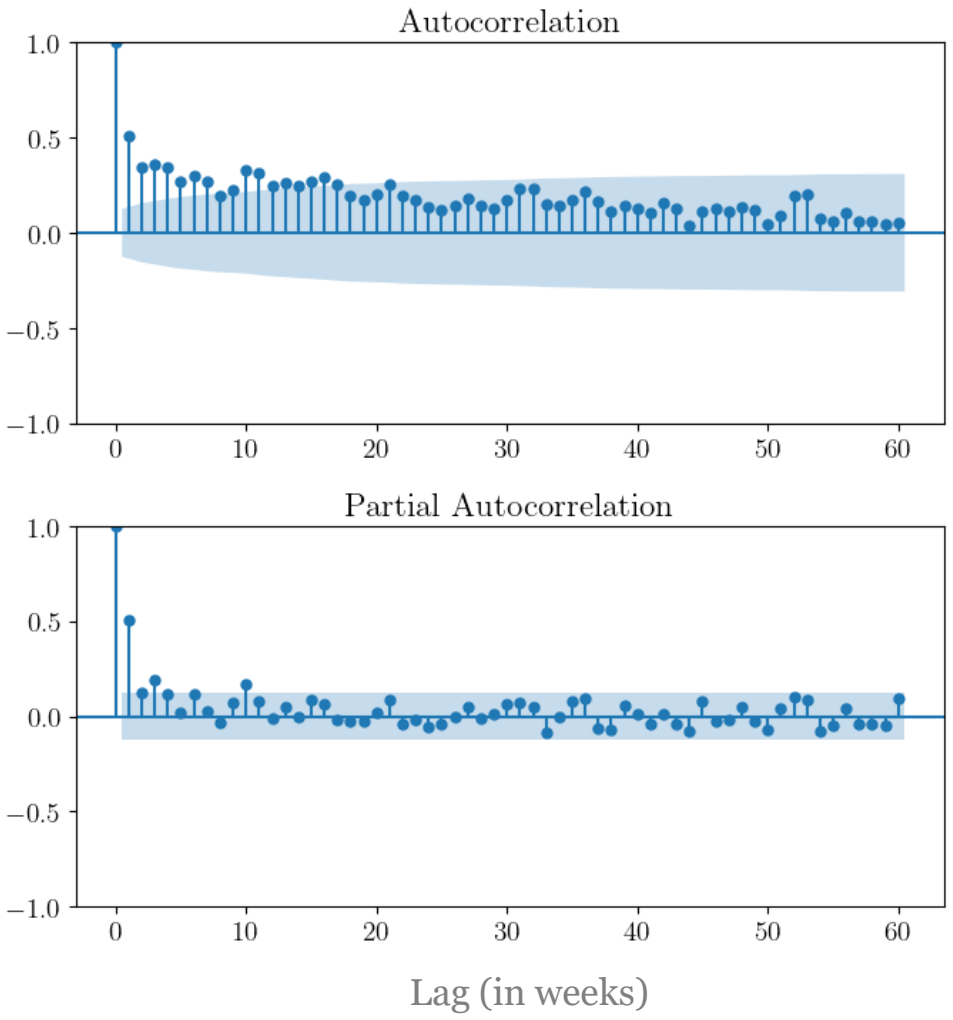
Direct correlation only — the indirect links are removed. Cleaner for spotting AR.



Sharp cutoff after lag 4 → set p = 4. Use PACF to guide p (AR order).

Applying Diagnostic Plots to Arrival Rates

- Returning to our arrival data, we can plot the autocorrelation and partial autocorrelation functions
- We see evidence of significant auto and partial autocorrelation lags < 10 days (1.4 week)
- The PACF cutting off near lag 1 suggests a relatively simple AR(1) process might be sufficient
  - The dominant signal is just "this week looks like last week”
  - The slow ACF decay is consistent with that



Alternative Model Fitting Method

- Another common way of fitting models is the Box-Jenkins Method
  - We define a set of p, d, q values we want to try fitting a model with (say 0-5 lags for each)
  - Repeatedly fit models for all combinations and select the model that minimizes the AIC
- AIC (Akaike Information Criterion):
  - Commonly used metric for measuring the fit of a model using its likelihood (L) and the number of fitted parameters (k). **Lower is better**

$$AIC = 2k - 2\ln(L)$$

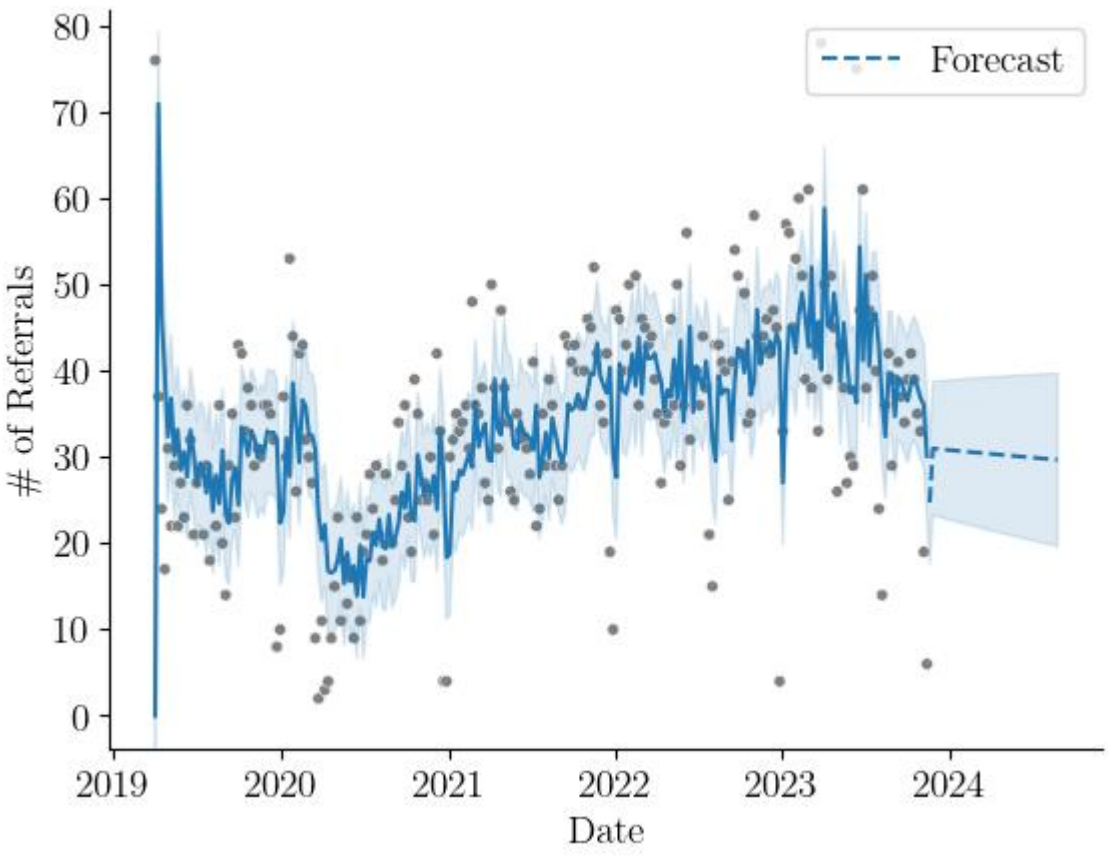
  - The 2k term penalizes complexity → adding parameters only helps if the fit improves enough to justify them

Lag (in weeks)

Arrival Rate Forecasting – ARIMA Model 1

- Using the Box-Jenkins method, we look for the model that achieve the lowest AIC
- Then, we can look at the best model's forecast of future arrivals

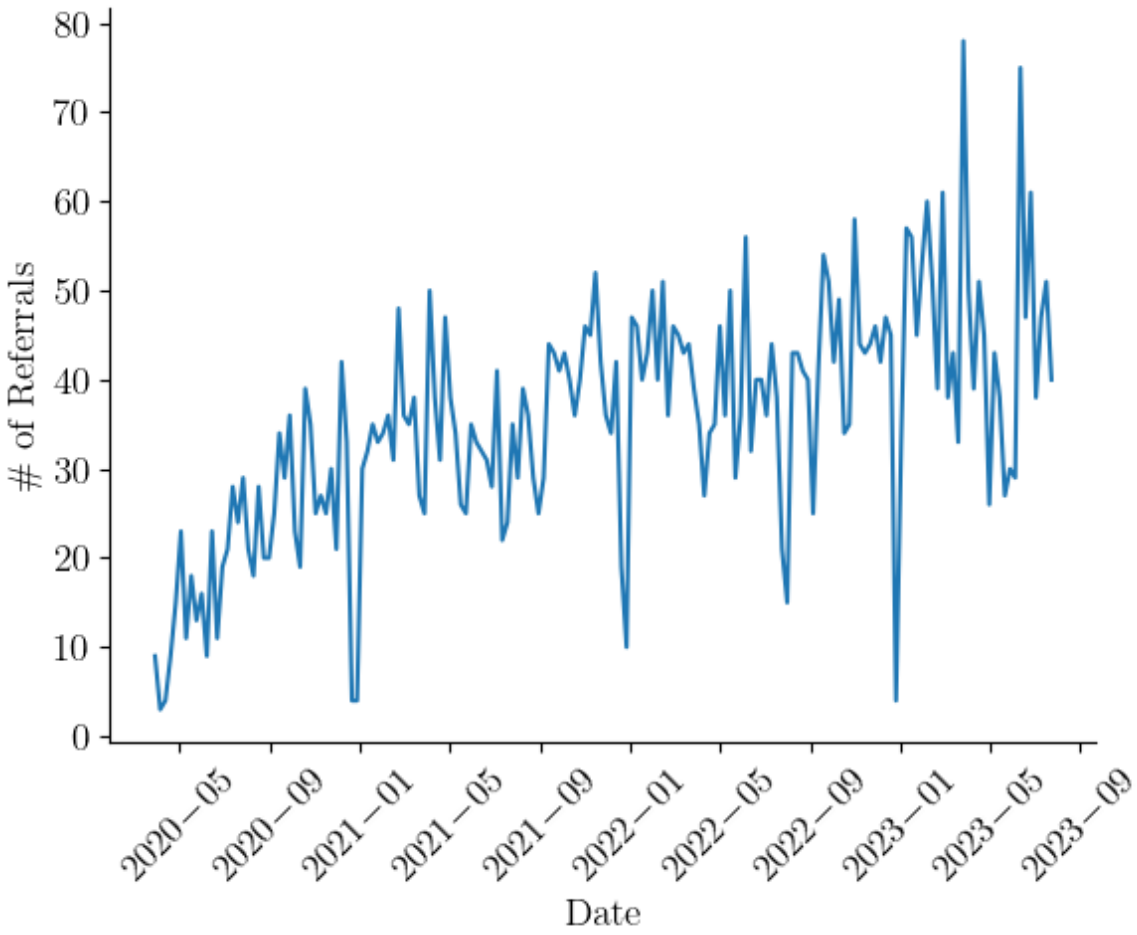
P / Q Terms	0	1	2	3	4	5
0		2215.4	2111.1	2066.2	2032.9	2001.8
1	1904.6	1853.9	**1842.5**	1844.4	1846.2	1848.8
2	1884.6	1842.8	1844.4	1846.2	1848.4	1849.6
3	1864.7	1844.6	1846.4	1848.1	1846.5	1847.6
4	1859.3	1845.1	1848.5	1844.9	1849.7	1846.6
5	1860.2	1846.9	1849.0	1850.2	1853.1	1851.6



- Chosen parameters: ARIMA (p=**2**, d=**1**, q=**1**)

Arrival Rate Forecasting – ARIMA Model 2

- We need to clean the time series
- Truncate the beginning and end of the time series that may be contaminated by:
  - System being brought online at the beginning of COVID
  - System being changed over in October 2023
- We observed mild non-stationarity in the time series
  - Referrals marginally increase over time



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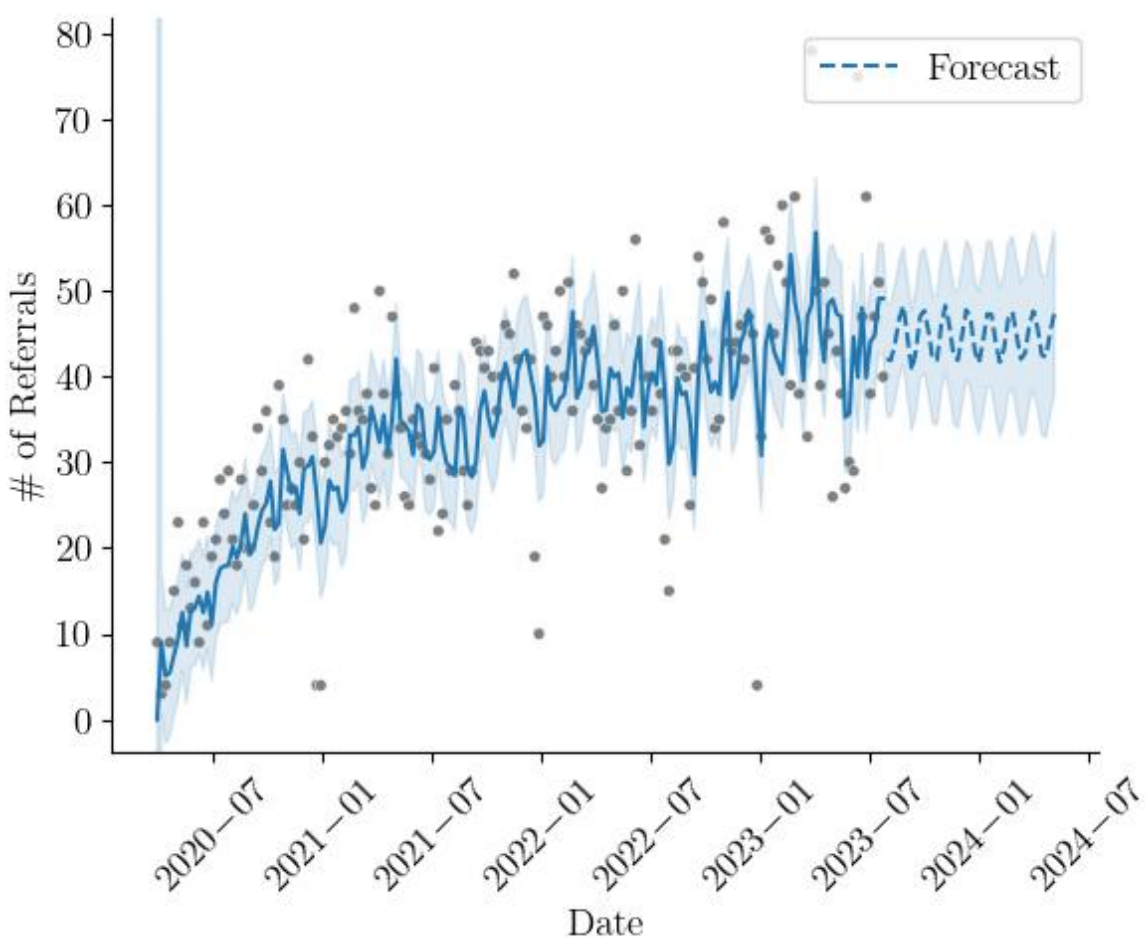
Arrival Rate Forecasting – ARIMA Model 2

- Consequently, the best fitting model on cleaned data is an ARIMA(4,1,5) process
  - p=4 → we look back 4 weeks
  - d=1 → one round of differencing
  - q=5 → corrects on 5 weeks of errors
- In this case, the forecast seems to be that the mean weekly referral number will be ~44.7
- The further out we forecast, the uncertainty (shaded blue band) grows



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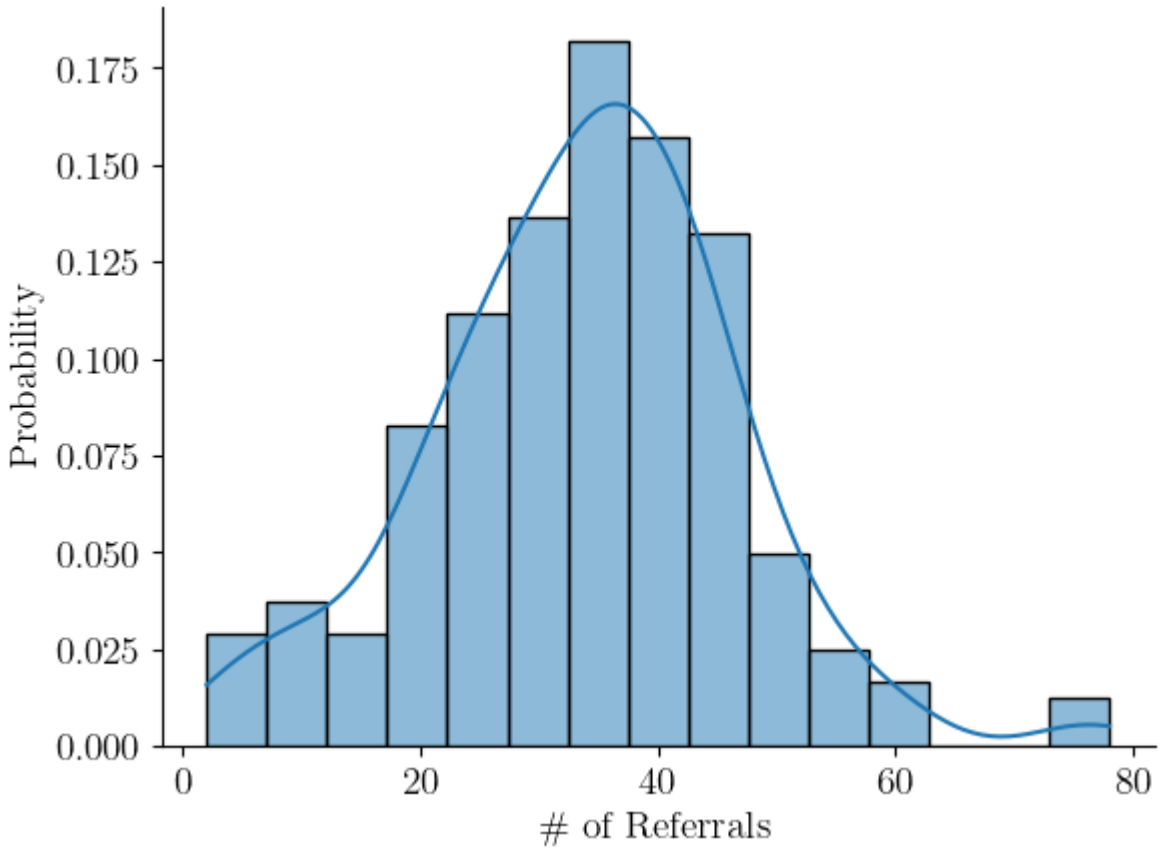
Predicting Future Arrivals – Modelling Implications

- Interestingly, arrivals are not Poisson distributed
- This has implications for queueing models as it violates a key assumption of many easily tractable models



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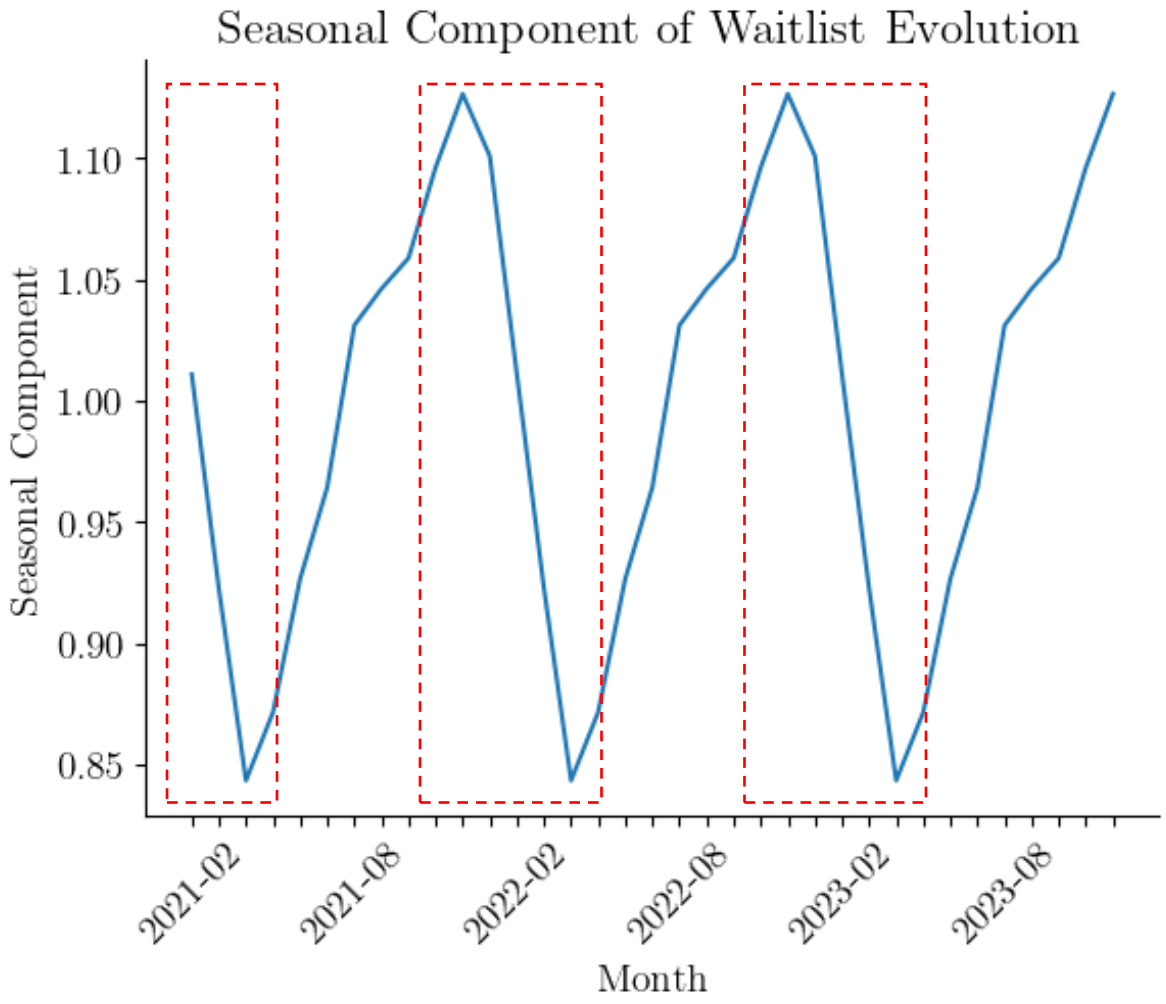
PREDICTING: VOLUME OF AGING-OUT PATIENTS

Using ARIMA with Seasonality

Predicting the Volume of Aging-Out Patients

- There is evidence of **seasonality** in the time series
- This is likely due to:
  - Rush to assess/service waiting patients that are older than 3 y.o. at the beginning of each year (before they soon turn 4 y.o.)
- We see clear evidence of this seasonality when we extract the seasonal component from the time series
- Knowing when surges will occur allows WestField to plan staffing, prioritize waitlist, and reduce the number of children who age out without services



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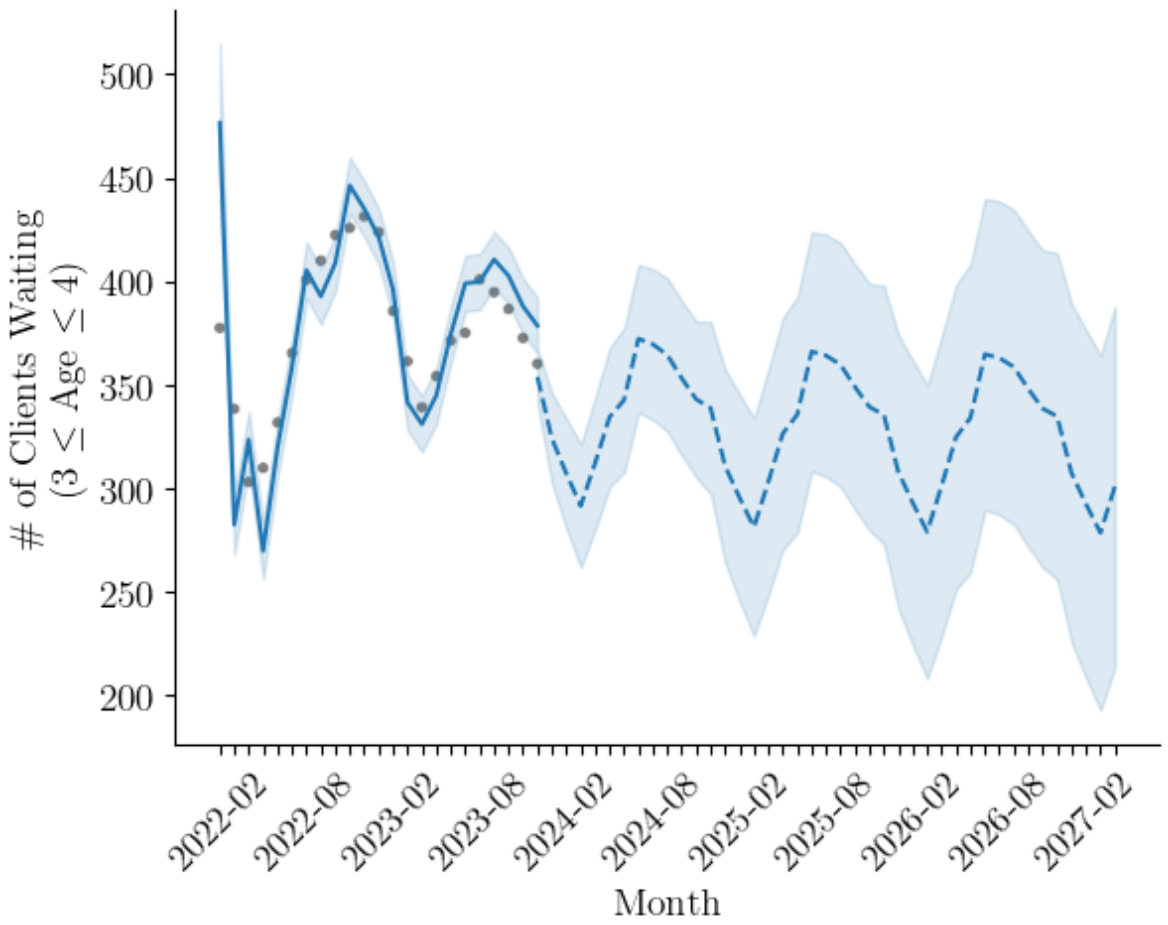
Forecasting Volume of Aging-Out Patients: ARIMA

- Incorporating a seasonality component, the best fitting ARIMA model is:
  - ARIMA(1,0,1) with a constant annual seasonality component
- We can see that the number of near-age-out patients spikes at the beginning of each year (likely due to the annual surge driven by families who rush to access these services while still eligible)
- The spike is usually ~150-200 children which is not a minor fluctuation
- The model also expects this seasonality to continue in the coming years



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PREDICTING: APPOINTMENT ATTENDANCE

Using Machine-Learning Methods

Attendance Prediction

- In many service settings, appointments have multiple outcomes:
  - Attendance
  - No-show (where a client misses their appointment without providing notice)
  - Cancellation (where a client provides some kind of advanced notice)
- In the case of WestField, the most impactful types of non-attendance are:
  - No-shows and cancellations within 24 hours of their appointment (last-minute cancellations)
- These appointments are extremely hard to impossible to replace with another client, leading to decreased effective usage of clinical capacity

Attendance Prediction

- Predicting what appointments are likely to result in a no-show or cancellation has multiple benefits:
  - Can incorporate these predictions into overbooking models
  - Can use causal models to understand what drives no-shows and cancellations in an effort to reduce their likelihood
  - Can use methods such as targeted reminders to reduce the likelihood of an appointment resulting in a no-show or last-minute cancellation

Journal of General Internal Medicine

Springer

J Gen Intern Med. 2023 Jun 20;38(Suppl 3):894–904. doi: 10.1007/s11606-023-08131-5

Using Nudges to Reduce Missed Appointments in Primary Care and Mental Health: a Pragmatic Trial

ACI Applied Clinical Informatics

Thieme

Appl Clin Inform. 2014 Sep 24;5(3):836–860. doi: 10.4338/ACI-2014-04-RA-0026

Patient No-Show Predictive Model Development using Multiple Data Sources for an Effective Overbooking Approach

Types of Predictive Models

Parametric

Assumption: Make assumptions about the underlying distribution of the data (e.g. relationships are linear)

Pros: Generally more powerful when assumptions hold; often more interpretable

Examples: Linear regression, logistic regression, multinomial logistic regression

Nonparametric

Assumption: Make no assumptions about the underlying distribution (learn patterns from scratch)

Pros: Often good at prediction regardless of whether distributional assumptions hold

Cons: Generally low interpretability (black-box); risk of overfitting to training data

Examples: Decision trees, random forests, XGBoost, neural networks

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Binary Logistic Regression – Will They Show Up?

- We want to predict if an appointment will be **Attended** or **Not Attended**
- Every feature features a single probability up or down

1 Collect features

Age, modality, history, distance, clinician, time of day, etc.

2 Multiply by weights

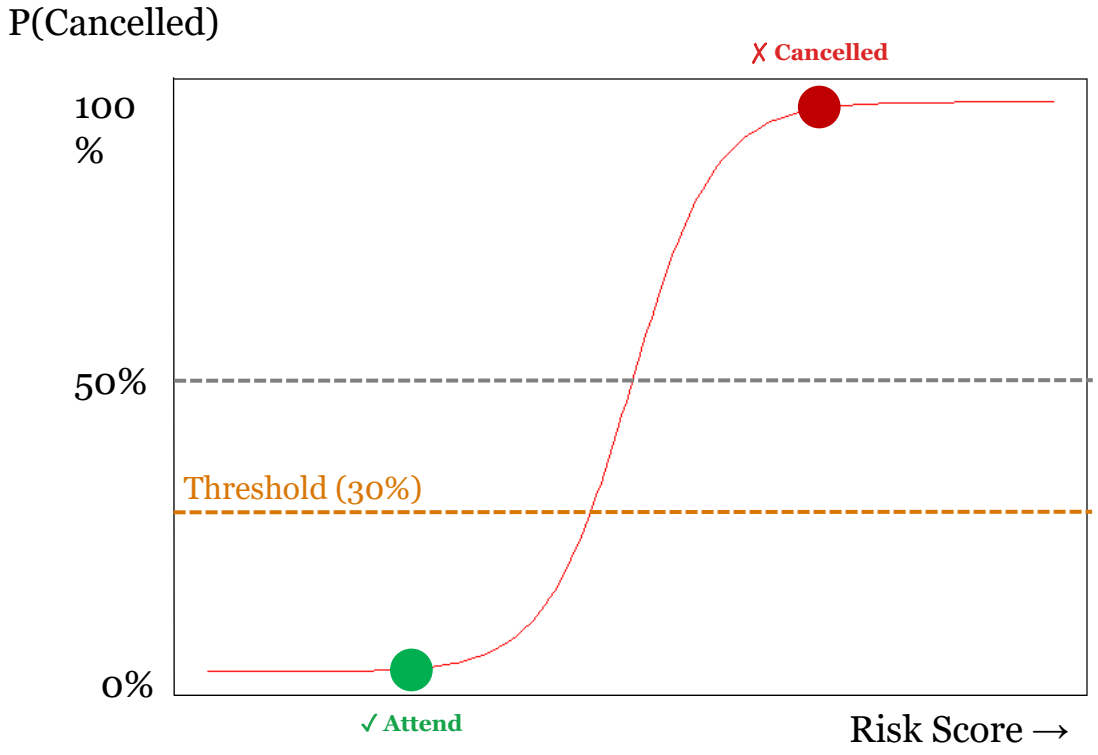
Each feature gets a learned weight (positive raises risk, negative lowers it)

3 Squash to 0–100%

The logistic (S-curve) function converts any score into a probability.

4 Apply a threshold

If P(cancelled) > 30% → flag as at-risk. Otherwise expect attendance. The threshold is adjustable: **lower it to catch more at-risk appointments; raise it to reduce false alarms**



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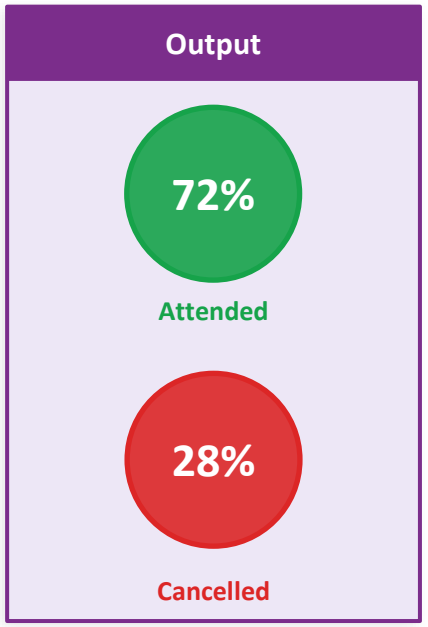
Binary Logistic Regression – Weighing the Features

- Each feature gets a weight. The model adds them up, applies the S-curve, and outputs one number to represent the probability of cancelling



Result: P(Cancelled) = 28% → below our 30% threshold → predict Attended

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Cancellation Data

- We have records of patient treatment timelines
  - Modality of the appointments (e.g., virtual or face-to-face, group)
  - Patient type (Scale of 1-3 where 3 = highest priority patient, 1 = lowest priority)
  - Whether the appointment was attended or not

Client ID	Patient Type	Appt. Modality	Attended (Y/N)	Appt. #
0	1	F	Y	1
0	1	F	Y	2
0	1	F	N	3
0	1	F	Y	4

Appointment Attendance Predictive Model 1

- We will start with a simple regression model. We will use a linear term for the patient type and a linear term for the modality

$$P(\text{Attended}) = \frac{e^{a+\beta_1(Patient\ Type)+\beta_2(Modality)}}{1+e^{a+\beta_1(Patient\ Type)+\beta_2(Modality)}}$$

Variable	Coef	Std Err	P > |z|
Intercept	0.945	0.167	0.000
C(pat_type)[T.2]	0.258	0.098	0.003
C(pat_type)[T.3]	0.551	0.204	0.007
C(modality)[T.G]	0.107	0.163	0.512
C(modality)[T.V]	1.1207	0.171	0.000

Baseline (Intercept) = Patient Type 1 and Modality Face-to-Face

The model's AUC on the test set is **0.641**.

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Appointment Attendance Predictive Model 2

- The second regression model will be more complex: a linear term for the patient type, a linear term for the modality, and an interaction term between them

$$\text{Attended} = \frac{e^{a+\beta_1(Patient\ Type)+\beta_2(Modality)+\beta_3(Patient\ Type \times Modality)}}{1+e^{a+\beta_1(Patient\ Type)+\beta_2(Modality)+\beta_3(Patient\ Type \times Modality)}}$$

Variable	Coef	P > |z|
Intercept	0.987	0.000
C(pat_type)[T.2]	0.195	0.001
C(pat_type)[T.3]	0.600	0.000
C(modality)[T.G]	0.008	0.891
C(modality)[T.V]	0.902	0.000
C(pat_type)[T.2]:C(modality)[T.G]	0.013	0.876
C(pat_type)[T.3]:C(modality)[T.G]	-0.065	0.429
C(pat_type)[T.2]:C(modality)[T.V]	0.332	0.003
C(pat_type)[T.3]:C(modality)[T.V]	0.738	0.000

The model's AUC on the test set is **0.640**.

This interaction term did not add more value to the model's predictive performance, so we can omit this complexity and use Model 1.

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Going from Coefficients to Probabilities

- Using Model 1 results, build the linear score (add up the intercept + the coefficients of the patient's characteristics):
  - **Score = intercept + β (pat_type) + β (modality)**
  - For example, a Type 2 patient with a virtual appointment:
    - Score = 0.945 + 0.258 + 1.207 = 2.410
- Then apply the logistic curve (S-curve) function:
  - P(Attended) = $\frac{e^{score}}{1+esc^{ore}}$ → $\frac{e\text{^}2.140}{1+e\text{^}2.140}$ = 0.918

Variable	Coefficient
Intercept	0.945
C(pat_type)[T.2]	0.258
C(pat_type)[T.3]	0.551
C(modality)[T.G]	0.107
C(modality)[T.V]	1.1207

	pat_type	modality	attend_prob_mod_1
0	1	F	0.720202
1	1	V	0.895882
2	1	G	0.716663
3	2	F	0.769198
4	2	V	0.917633
5	2	G	0.766077
6	3	F	0.833136
7	3	V	0.943473
8	3	G	0.830690

We can now use these average probabilities of attendance of each patient class + modality combination directly as parameters in a linear program (to be used in later lectures)

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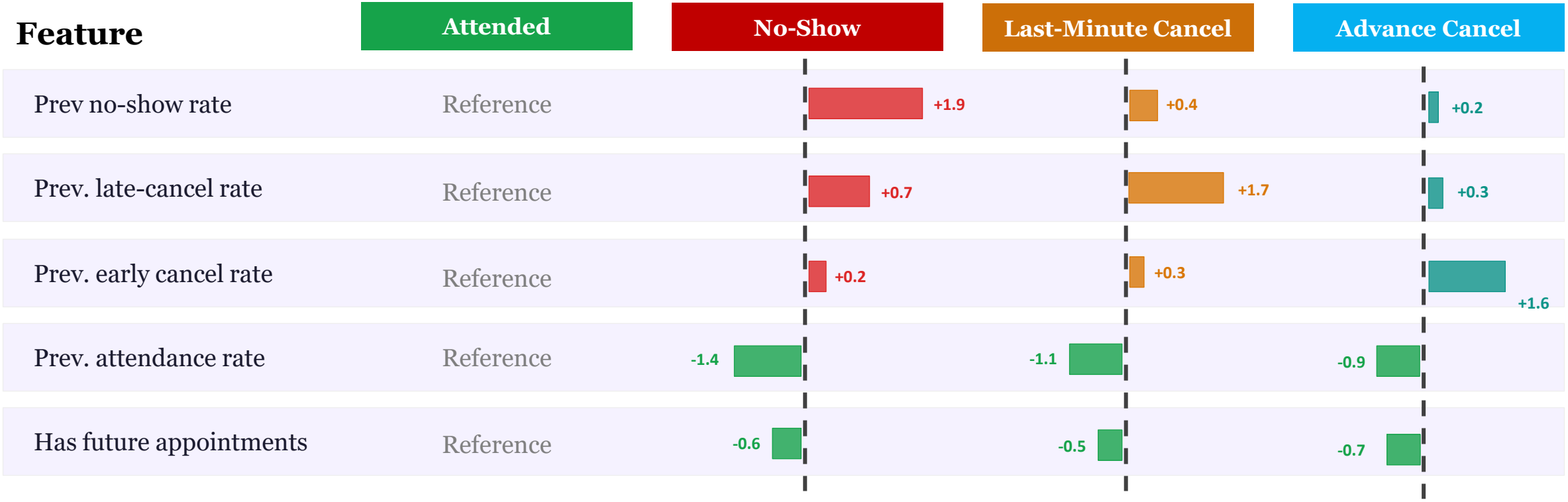
Multinomial Logistic Regression – One Equation Per Outcome

- What if we want to predict more than if an appointment is attended or cancelled. We want to know what type of cancellation it would be: **No-Show**, **Last-Minute Cancel**, or **Advance Cancel** (or **Attended**)



Each non-reference class gets its own set of weights vs. the reference. "Attended" is the reference so all other coefficients express change relative to it → P(Attended) = 1 – P(No-Show) – P(Late Cancel) – P (Advance Cancel)

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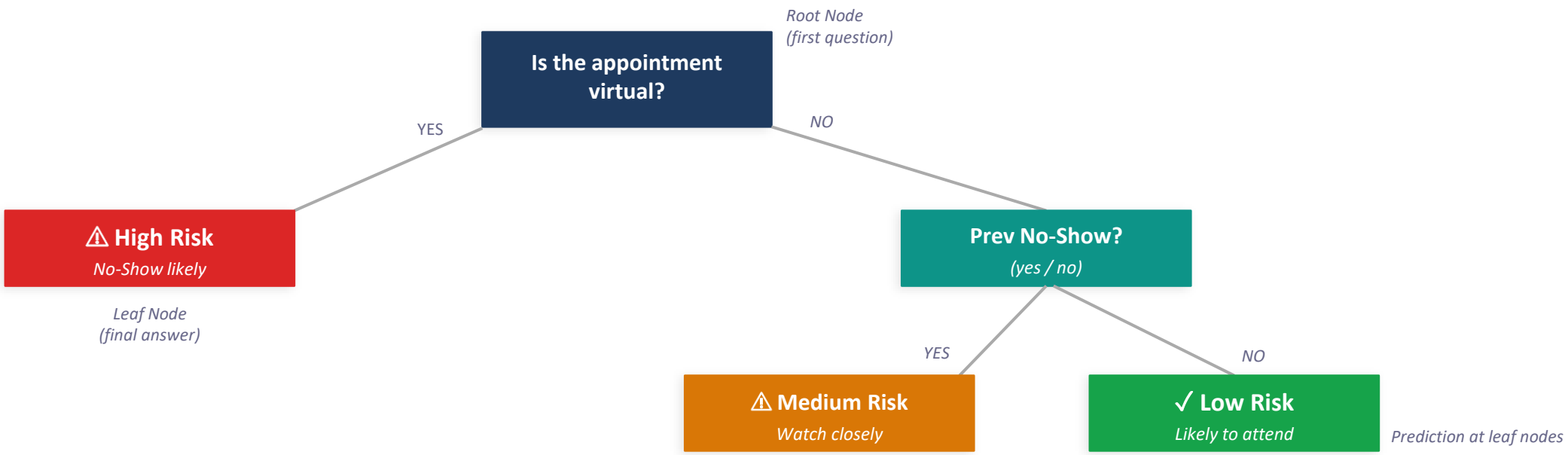
Decision Trees (Nonparametric Model)

- Asks a series of yes/no questions about an appointment. Each question narrows down the most likely outcome.

Example:



How we build it:

1 All appointments start together **2** Find the question with highest **Information Gain**
3 Split, then repeat in each branch **4** Stop when groups are pure enough

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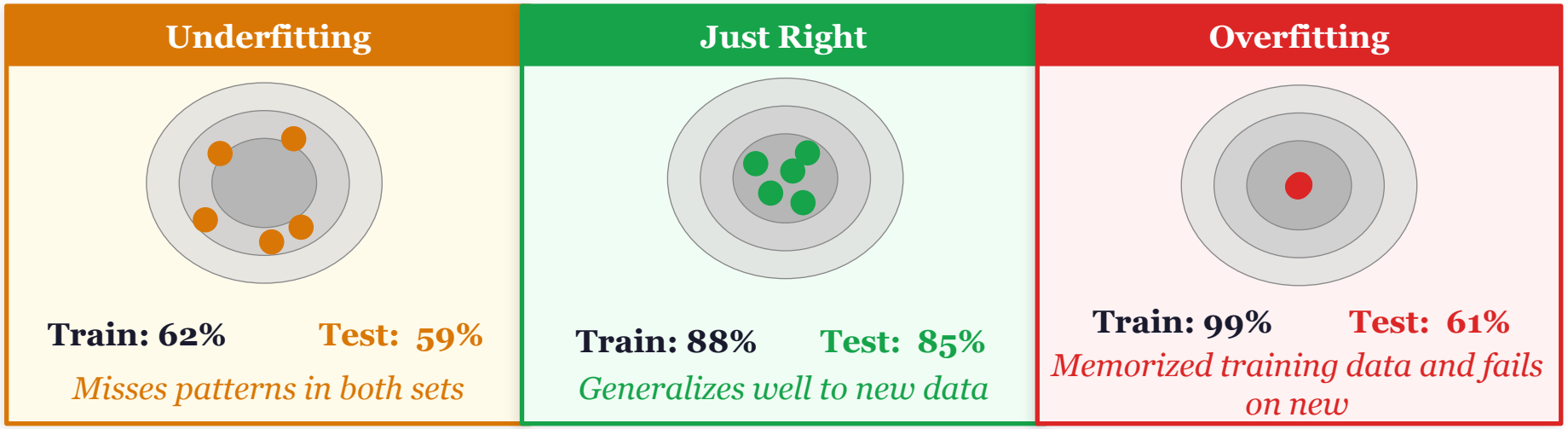
Overfitting – When a Model Memorizes Instead of Learns

- A model that's too complex memorizes the training data (a disadvantage of nonparametric models) which makes it perform well on data its seen but poorly on new data.



- How to fix this? **Ensemble Methods**
  - **Random Forest**: averages many decision trees trained on sliced samples over the overall dataset → minimizes variance and overfitting
  - **XGBoost**: builds trees that correct each other's errors → minimizes bias and underfitting

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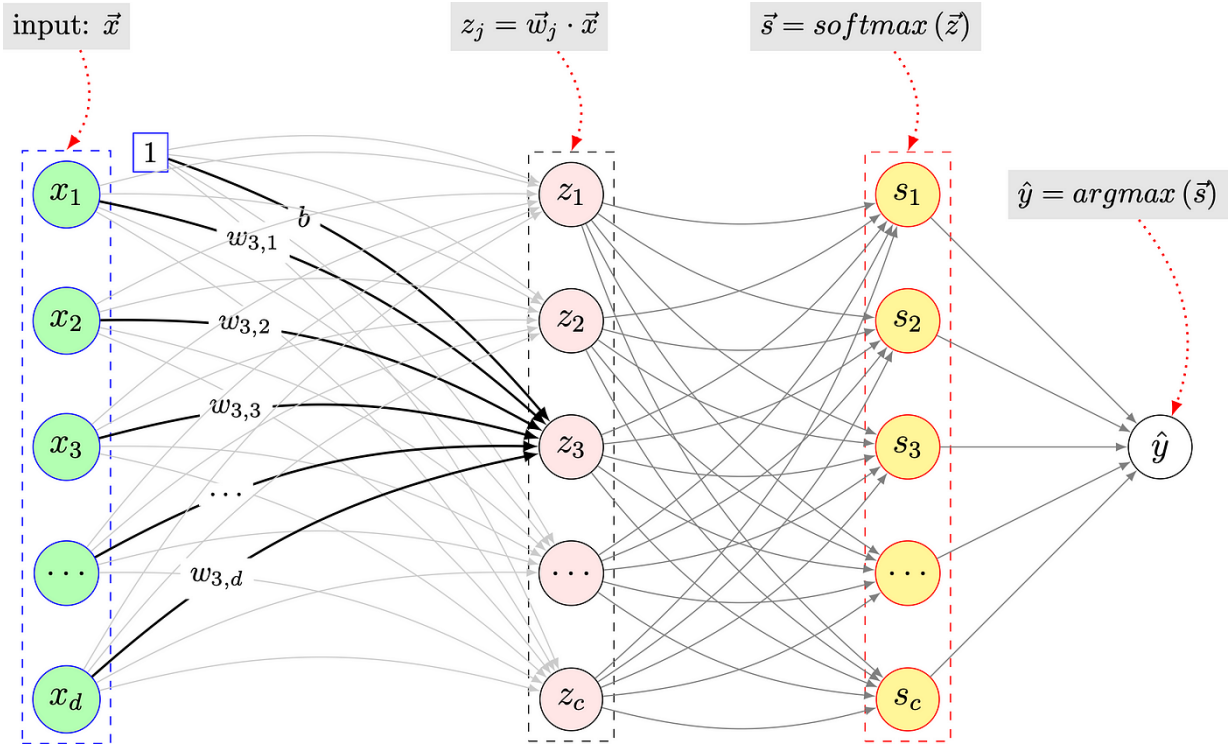
Neural Networks

- Passes data through layers of simple calculations. Each layer extracts slightly more abstract patterns. The final layer outputs a probability for each outcome.
- **Why neural networks are powerful:**
  - Each hidden layer learns increasingly abstract combinations of features. The network can model any relationship including ones that no single rule could capture.
  - But the downside is that this makes the learned weights uninterpretable



Model Performance Comparison

ROC AUC Scores (Average Across 5-fold Cross Validation)

Model	Attended	No-Show	Cancellation (w/in 24 Hrs)	Cancellation (> 24 Hrs)
Logistic Regression	0.830	0.905	0.878	0.908
Neural Network	0.906	0.966	0.931	0.965
XGBoost	0.921	0.970	0.937	0.972

What is ROC AUC? A score from 0.5 (random guessing) to 1.0 (perfect prediction). A score of 0.90 means the model correctly ranks a no-show appointment above an attended one 90% of the time. Higher is always better.

Logistic Regression

✓ Best interpretability: you can explain exactly why any appointment was flagged as high-risk. Ideal for communicating results to clinical staff.

✗ Lower accuracy — may miss nuanced patterns in data.

Neural Network

✓ Handles complex, non-linear relationships and interactions between features automatically.

✗ Black box: difficult to explain specific predictions. Also prone to overfitting.

XGBoost

✓ Best overall accuracy with reasonable interpretability via feature importance scores. Strong balance of performance and explainability.

✗ Less interpretable than logistic regression.

PREDICTING: NUMBER OF APPOINTMENTS

Using Linear Regression

Number of Appointments Predictive Model

- Given the continuous nature of the number of appointments needed in this setting we are going to use a linear regression model (OLS)
  - This hinges upon every patient being treated to completion (i.e., discharged due to achieving the desired outcomes), which is not always the case in reality for WestField
- We are interested in the number of treatments a patient needs to complete service
  - This is different from the total number of bookings they had as some of these appointments may have been cancelled
    - $N_{needed} \neq N_{booked}$
- Thus, we filter out the cancelled appointments and recompute the total number of appointments per patient

Client ID	Patient Type	Appt. Modality	Attended (Y/N)	Appt. #
0	1	F	Y	1
0	1	F	Y	2
0	1	F	N	3
0	1	F	Y	4

Client ID	Patient Type	Appt. Modality	Attended (Y/N)	Appt. #
0	1	F	Y	1
0	1	F	Y	2
0	1	F	Y	3

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Number of Appointments Predictive Model 1

- The first regression model will be simple: a linear term for the patient type and a linear term for the modality $N = a + \beta_1(Patient\ Type) + \beta_2(Modality)$

OLS Regression Results

Dep. Variable:	num_appt	R-squared:	0.789
Model:	OLS	Adj. R-squared:	0.788
Method:	Least Squares	F-statistic:	1676.
Date:	Mon, 03 Mar 2025	Prob (F-statistic):	0.00
Time:	20:53:17	Log-Likelihood:	-3763.2
No. Observations:	1800	AIC:	7536.
Df Residuals:	1795	BIC:	7564.
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	8.4467	0.103	81.755	0.000	8.244	8.649
C(pat_type)[T.2]	2.9083	0.113	25.697	0.000	2.686	3.130
C(pat_type)[T.3]	7.7517	0.113	68.491	0.000	7.530	7.974
C(modality)[T.G]	4.6717	0.113	41.277	0.000	4.450	4.894
C(modality)[T.V]	0.9033	0.113	7.982	0.000	0.681	1.125

Omnibus:	27.143	Durbin-Watson:	1.727
Prob(Omnibus):	0.000	Jarque-Bera (JB):	25.264
Skew:	0.248	Prob(JB):	3.27e-06
Kurtosis:	2.699	Cond. No.	4.44

The model R^2 is quite high, indicating a good fit.

The intercept term corresponds to Class 1 patients who receive in-person treatment.

We see that this patient type + modality pairing requires the fewest visits.

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Number of Appointments Predictive Model 2

- The second regression model will be more complex: a linear term for the patient type, a linear term for the modality, and an interaction term between them

OLS Regression Results

Dep. Variable:	num_appt	R-squared:	0.942
Model:	OLS	Adj. R-squared:	0.942
Method:	Least Squares	F-statistic:	3630.
Date:	Mon, 03 Mar 2025	Prob (F-statistic):	0.00
Time:	20:53:17	Log-Likelihood:	-2601.7
No. Observations:	1800	AIC:	5221.
Df Residuals:	1791	BIC:	5271.
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	10.0050	0.073	137.450	0.000	9.862	10.148
C(pat_type)[T.2]	1.9900	0.103	19.332	0.000	1.788	2.192
C(pat_type)[T.3]	3.9950	0.103	38.809	0.000	3.793	4.197
C(modality)[T.G]	0.9200	0.103	8.937	0.000	0.718	1.122
C(modality)[T.V]	-0.0200	0.103	-0.194	0.846	-0.222	0.182
C(pat_type)[T.2]:C(modality)[T.G]	2.1300	0.146	14.631	0.000	1.844	2.416
C(pat_type)[T.3]:C(modality)[T.G]	9.1250	0.146	62.680	0.000	8.839	9.411
C(pat_type)[T.2]:C(modality)[T.V]	0.6250	0.146	4.293	0.000	0.339	0.911
C(pat_type)[T.3]:C(modality)[T.V]	2.1450	0.146	14.734	0.000	1.859	2.431

Omnibus:	1.089	Durbin-Watson:	1.897
Prob(Omnibus):	0.580	Jarque-Bera (JB):	1.133
Skew:	0.058	Prob(JB):	0.568
Kurtosis:	2.957	Cond. No.	13.9

We see a higher R^2 (0.94 vs. 0.79). This tells us that the interaction factor is significant here (e.g., a Type 3 patient, who typically needs many more appointments, might respond very differently to virtual vs. in-person care than a Type 1 patient)

Interestingly, we see that virtual appointments do not have a significant effect on the efficacy of treatment for Class 1 patients

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From Predictions to Action

- Each prediction we built produces a parameter that can be used directly into Prescriptive Analytics

Predictive Output	Models Used	How It Feeds Prescriptive Analytics
Future weekly arrival rate	ARIMA (4,1,5)	Capacity planning: sets the λ (arrival rate) input for queueing models; triggers staffing adjustments before demand spikes hit.
Volume of aging-out patients per quarter	Seasonal ARIMA (1,0,1)	Waitlist triage: signals when to pre-emptively prioritize children near aging-out; informs targeted outreach to reduce avoidable age-outs.
P(Attended | patient type, modality) and type of cancellation	Logistic Regression (Model 1) + XGBoost	Attendance probabilities can be used in building overbooking models or scheduling policies.
Expected # of appointments per patient (by type and modality)	OLS Regression (Model 2)	Episode capacity allocation: predicted treatment length sets how much future capacity each new intake 'reserves', enabling realistic throughput planning

What's Next?

- So far, we have analyzed and made predictions using tabular, structured data
- However, WestField's clinical records also contain a large volume of free-text: referral notes written by physicians, therapist progress notes, and discharge summaries
  - These likely contain some of the richest signals about what's happening and more importantly, **why it's happening**
- Consider what structured data cannot explain: two patients with identical type codes, age, and modality assignments may have vastly different trajectories → one discharges in 8 visits, another requires 20 visits before discharge
- The challenge is that free-text is harder to analyze systematically, so we will explore methods for data mining unstructured clinical text in upcoming lectures